Electronic structure calculations: draft main-text section

1 Computational results and their relation to the observed dynamics

Electronic structure calculations were performed to identify the states reached at the two excitation wavelengths and to determine which nuclear coordinates they drive. All calculations used the TPSSh functional with a mixed def2-TZVP(Fe,O)/def2-SVP(C,H) basis and SMD aqueous solvation, matching the solvent of the experiment; the ground state was optimised without symmetry constraints and confirmed as a true minimum. Excited states were obtained from TDA-TD-DFT with 80 roots, and the quartet manifold from spin-flip TD-DFT with 140 roots. Full details, tables and the complete mode-by-mode analysis are given in the Supporting Information.

1.1 Both excitation windows access ligand-to-metal charge transfer

The calculated spectrum places a state at 2.92 eV (425 nm) and a second at 4.52 eV (277 nm), matching the two experimental excitation windows. Both are dominated by $\beta \to \beta$ excitations into the vacant Fe(III) β-d manifold and both retain sextet spin ($\langle S^2 \rangle = 8.80$ and 8.96 against 8.75 for a pure sextet), so both are ligand-to-metal charge transfer in character.

The two are nevertheless distinguished by the nuclear coordinate they drive. Projecting each excited-state gradient, evaluated at the ground-state geometry, onto the ground-state normal modes gives the displacement each mode undergoes on vertical excitation. For the lower state the largest activity is the conjugated C=C/C=O ring stretch at 1560 cm^{-1}; for the higher state it is the Fe–O chelate deformation at 435 cm^{-1}, with a Huang–Rhys factor of 1.94 against 0.80. The higher-energy excitation distorts the metal–ligand framework two to three times more strongly, and deposits nearly twice the total reorganisation energy (0.59 against 0.32 eV). The computed Fe–O mode at 435 cm^{-1} corresponds to the band observed near 449 cm^{-1}.

1.2 Why the same three modes are populated at both wavelengths

The reorganisation energy is concentrated in modes that the measurement does not monitor. Summed over the ν_8, ν_9 and ν_{10} windows, the Huang–Rhys factors amount to $S \approx 0.01$–0.04, and these bands together carry only 2–6% of the total resonance-Raman activity. The dominant Franck–Condon activity lies instead at 150–450 cm^{-1} and at 1560 cm^{-1}.

The three observed bands are therefore not the primary distortion coordinates of either excited state. They are reporters of the vibrational bath, populated by whatever relaxation cascade is running above them. This accounts for the observation that ν_8, ν_9 and ν_{10} appear with the same prompt rise at both excitation wavelengths despite the two states differing by 1.6 eV in energy and by a factor of two in reorganisation energy: the bands report the arrival of vibrational energy rather than the identity of the state that delivered it.

1.3 The 941 cm^{-1} transient band and excited-state symmetry breaking

A charge-transfer excitation in a tris-chelate must lift the degeneracy of the doubly degenerate ligand vibrations. In D_3 the ligand-framework modes transform as A_1, A_2 and E, and the three equivalent chelate rings give tight manifolds: the calculated ν_8 region spans only 3.4 cm^{-1}, with an A–E separation of 1.3 cm^{-1}. A charge-transfer hole must be accommodated by three equivalent rings, and wherever vibronic coupling exceeds that small intrinsic splitting the delocalised arrangement is unstable and the hole localises on one ring, as valence localises in a mixed-valence compound. The distortion that achieves this destroys the equivalence on which the E degeneracy rests.

The calculations confirm this on two independent grounds. Structurally, the six Fe–O distances span 0.014 Å in the ground state but 0.098 and 0.241 Å in the two excited states, the lower state showing two short and four long distances, the signature of a single ring singled out. Vibrationally, all twelve near-degenerate ground-state manifolds between 200 and 1700 cm^{-1} widen in both excited states without exception, by median factors of 13 and 15 and by no less than three. The ν_8 manifold widens from 3.4 to 57 cm^{-1}, by factors of 16.7 and 16.6 — the same at both wavelengths to within 1 cm^{-1}, confirming that the effect follows from charge-transfer character rather than from the properties of either state.

This bears directly on the transient band at 941 cm^{-1}, which appears at both excitation wavelengths and which lies 63 cm^{-1} above the ground-state ν_8 position. The calculations do not reproduce a rigid 63 cm^{-1} stiffening of ν_8: the individual descendants of that manifold shift by between −43 and +24 cm^{-1}. What they predict instead is that the manifold ceases to be a single narrow feature and spreads over some 57 cm^{-1}, so that the transient ν_8 region should be substantially broader than the ground-state doublet at 878 and 896 cm^{-1} and may resolve into additional components. Since the calculation already underestimates the ground-state splitting by a factor of five (Supporting Information), the true excited-state spread is likely larger still. On this reading the 941 cm^{-1} feature is best understood as a resolved component of a split ν_8 manifold rather than as a uniformly stiffened mode, which also accounts for its appearance at both wavelengths.

1.4 Why vibrational cooling is faster following 400 nm excitation

The measured cooling times differ by a factor of four to five between the two excitation wavelengths. This ordering does not follow from the properties of either state considered alone; if anything those properties suggest the opposite, since the higher state places more than twice the reorganisation energy into the low-frequency skeletal modes that couple most efficiently to the solvent. The difference instead reflects how many electronic states lie below each excitation.

Table 1: Electronic states below each excitation, from 80 sextet roots and 140 quartet roots computed at the ground-state geometry.

	2.92 eV (425 nm)	4.52 eV (277 nm)
Sextet states below	14	60
Quartet states below	17	59
Total	**31**	**119**
Nearest quartet / cm^{-1}	250	121
Reorganisation energy / eV	0.32	0.59

The higher excitation has four times as many states beneath it, and the measured cooling times differ by four to five. We propose that the measured τ_2 reflects the duration of the electronic cascade rather than the intrinsic rate of energy transfer to the solvent. Each internal-conversion or intersystem-crossing step converts an electronic gap into vibrational quanta, so while the cascade is still descending it continuously resupplies vibrational energy that the solvent is simultaneously removing. Following 400 nm excitation 2.9 eV is released through some 31 states and the cascade is short, so the measured decay approaches the true cooling time. Following 270 nm excitation 4.5 eV is released through some 119 states, the cascade outlasts the cooling, and the observed decay is set by the cascade.

The quartet manifold supplies a further asymmetry. The lower state lies in a sparse region of the sextet manifold, 1049 cm^{-1} from the nearest same-spin state, but only 250 cm^{-1} from a quartet — comparable to k_BT and small against the Fe(III) spin–orbit coupling constant. Same-spin internal conversion is slow there and intersystem crossing should dominate, and because the quartet manifold extends below the lowest sextet, a single spin-flip step transfers the system onto a ladder that reaches further down than the sextet manifold does. The higher state couples to the quartets more strongly still, 121 cm^{-1} to the nearest, but gains nothing by it: at 4.52 eV it retains 59 quartets beneath it. The shortcut is only useful to a state already near the bottom of the manifold, and this is consistent with the faster sub-picosecond decay of the 941 cm^{-1} feature observed following 400 nm excitation.

This explanation is a statistical one, resting on computed densities of states rather than on calculated rates, and we present it as such. A quantitative treatment would require spin–orbit couplings between the relevant sextet and quartet pairs, from which intersystem-crossing rates could be evaluated using the reorganisation energies reported here.